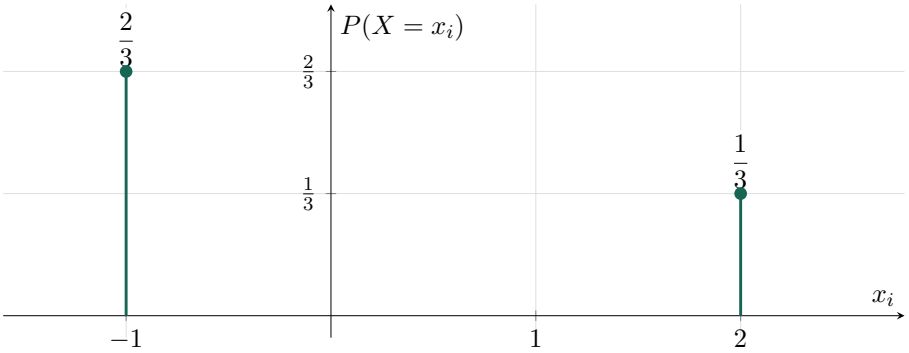
8.1 Variable, event and distribution

Definition and results

On a finite sample space Omega, a real random variable X is a function that assigns a real number to each outcome. The event $\{X = a\}$ groups the outcomes giving the value a. Its distribution assigns to each distinct value x_i probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Status and justification. Definitions. The events associated with distinct values form a partition of the sample space; this justifies adding the probabilities.

Worked example. A fair die gives a gain X equal to 2 for 1 or 2, and to -1 otherwise. The distribution is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3; 4; 5; 6\}$: probability $2\frac{}{3}$.



The height indicates the probability: $P(X = -1) = \frac{2}{3}$ & $P(X = 2) = \frac{1}{3}$. The exact probabilities appear on both bars.

Watch out

The values are not always equally likely even if the outcomes are. Group the outcomes that give the same gain.

Immediate application. Two fair coins are tossed. X counts the tails. Write its distribution and $P(X \leq 1)$.

8.2 Expectation and linearity

Definition and results

The expectation is $E(X) = \sum_i x_i p_i$. It describes the theoretical long-run mean, not an outcome necessarily possible on a single trial. For variables X and Y on the same finite sample space and real numbers a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, no assumption of independence.

Status and justification. Summing over the outcomes ω, $E(X + Y) = \sum_\omega (X(\omega) + Y(\omega))P(\omega)$; distributing the sum gives the result. The constant b has expectation b because the probabilities sum to 1.

Worked example. For the previous distribution $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$: the game is fair if X is the net gain. For $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

Watch out

An expectation is not an individual guarantee; a game can lose over every time span observed despite a positive expectation.

Immediate application. A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

8.3 Variance, standard deviation and König·Huygens

Definition and results

Denoting $\mu = E(X)$, $V(X) = E((X - \mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$. The König-Huygens formula gives $V(X) = E(X^2) - E(X)^2$. A variance is always positive or zero; the standard deviation has the same unit as X, the variance a squared unit.

Status and justification. Expand ($X - \mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X) = \mu$: $E(X^2) - 2\mu^2 + \mu^2$. We get the formula.

Worked example. For X with values 0 and 4 with probabilities $\frac{3}{4}$ & $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0 - 1)^2(\frac{3}{4}) + (4 - 1)^2(\frac{1}{4}) = 3$.

Watch out

$E(X^2)$ is not $E(X)^2$. A negative computed variance signals an error or a very small rounding error to be dealt with explicitly.

Immediate application. For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

8.4 Transforming and comparing gains

Definition and results

For real a, b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$. A translation moves the mean without changing the spread; a multiplication changes the scale. Two games with the same expectation can have very different spreads.

Status and justification. $aX+b-E(aX+b) = a(X-E(X))$. Squaring and then taking the expectation, we get $a^2V(X)$. The square root of a^2 is $|a|$.

Worked example. Game A always pays 2. Game B pays 0 or 4 with equal probability. The two have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same mean.

Watch out

The variance of a sum is not, in general, the sum of the variances; no rule of this kind is used without hypotheses.

Immediate application. $E(X) = 3$, $V(X)$ = 4. Give the parameters of $Y = -2X + 5$.

8.5 Samples and the observed mean

Definition and results

Here, a sample of size n is a list of n independent realisations of the same variable X. Its observed mean m varies from one sample to another. We compare m with $\mu = E(X)$; repetitions make it possible to observe the proportion of samples satisfying $|m-\mu| \leq \frac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task in the curriculum, not a claim of an exact rate.

Status and justification. Experimentation: neither a convergence theorem nor the normal distribution is proved here. We check list sizes, the mean calculation and the distinction between n and the number N of samples.

Worked example. For X with equally likely values 0 and 2, $\mu = 1$, σ = 1. For n=100, threshold $\frac{2}{10} = 0{,}2$: count the means in [0 ,8; 1,2]. A result of 956 out of 1000 means an observed frequency of 0.956 for this simulation only.

Watch out

Do not copy a single draw n times; do not state that the proportion is necessarily 95 %.

Immediate application. For $\mu = 5$, σ = 3 and n=36, give the requested experimental interval.